Run report: `f4_fp` before and after, against msolve and Singular, run-20261011T2359Z

Observations only. Interpretation: `../../REPORT.pdf`.

What ran

- **Runner:** `research/f4_fp_msolve_parity_20261011/run.sh <export_main> <export_new> <this dir> 3 300`.
- **Arms:** release builds of `examples/gb_cas_export.rs` with `src/cryptanalysis/f4_fp.rs` as on `main` (`4b6e09b78`, "main") and as on branch `f4-fp-msolve-gap` (`cba784a41`, "new"); binary SHA-256 in `raw/versions.txt`. Each at `RAYON_NUM_THREADS` 1 and 4. msolve 0.6.5 `-g 2` (reduced basis) at `-t 1` and `-t 4`, timed as process wall time (start-up included, about 6–10 ms). Singular 4.3.2 `std` under `dp/redSB`, timed inside Singular, 300 s limit.
- **Workload:** the 13 systems of `examples/f4_fp_bench.rs` (tier `fp`) and quad_n10, quad_n11, cubic_n6 (tier `fp-xl`): random dense systems, fixed seeds, grevlex.
- **Repetitions:** 3 interleaved rounds. Inside a round the `f4_fp` arms report the median of 3 repetitions (`fp`) or 1 (`fp-xl`).
- **Not run:** the main arm on quad_n11. On this host its quad_n10 takes 131 s single-threaded, so quad_n11 would take many minutes per repetition. Singular on quad_n11 hit the 300 s limit in all 3 rounds.
- **Host:** Claude Code cloud container, `Intel Xeon @ 2.80GHz`, 4 vCPU, shared, no pinning (isolation L0). An earlier attempt on 2026-10-11 (22:37Z) was lost to a container restart and is not used.

Medians (ms) and agreement

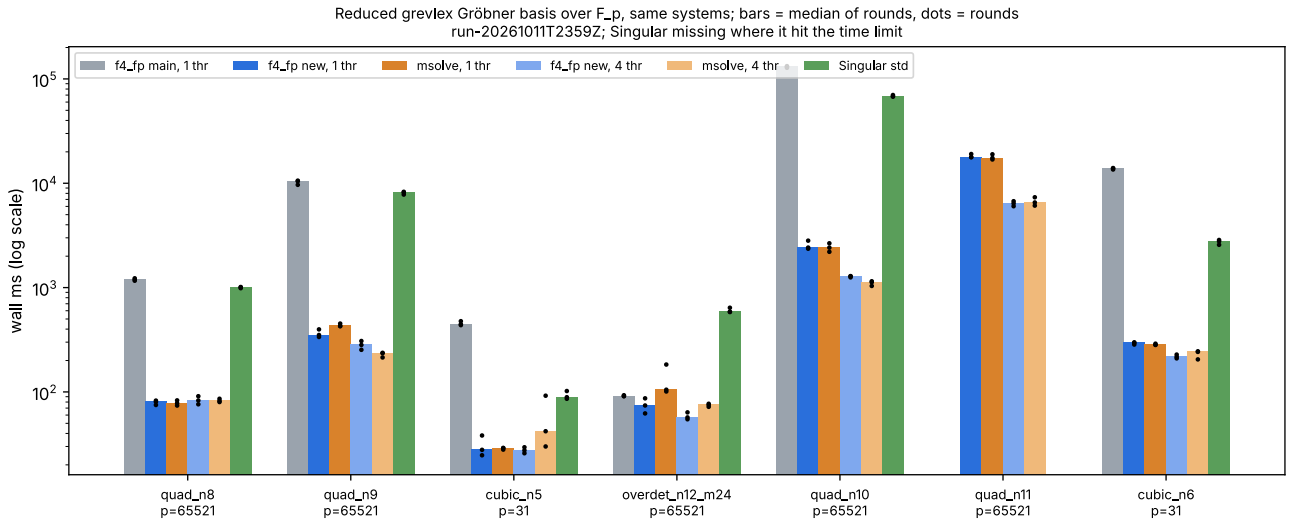


Figure 1: Larger systems, run-20261011T2359Z; bars at the median of 3 rounds, dots per round.

case	main 1	new 1	msolve 1	main 4	new 4	msolve 4	Singular
quad_n8	1195.2	80.2	78	780.3	82.8	82	1005
quad_n9	10463.8	351.3	429	5332.4	282.2	235	8165
cubic_n5	445.3	27.9	29	353.8	27.5	42	89
overdet_n10_m15	62.2	39.9	50	45.5	38.3	50	450
overdet_n12_m24	91.3	74.1	105	79.0	57.0	75	588
quad_n10	130759.5	2417.7	2417	51791.8	1285.6	1125	68360
quad_n11	not run	17795.2	17466	not run	6409.1	6544	> 300 s
cubic_n6	13864.8	295.1	283	6510.8	217.2	242	2778

All values measured; the remaining small systems are in `medians.json`. Basis fingerprints are identical between main and new on every system. In every round Singular found the new build's basis to be its own reduced Gröbner basis (45 of 45 checks). On quad_n11, where Singular timed out, Singular compared it with msolve's reduced basis: 767 elements each, each reducing to zero against the other (`raw/quad_n11_basis_check.txt`).

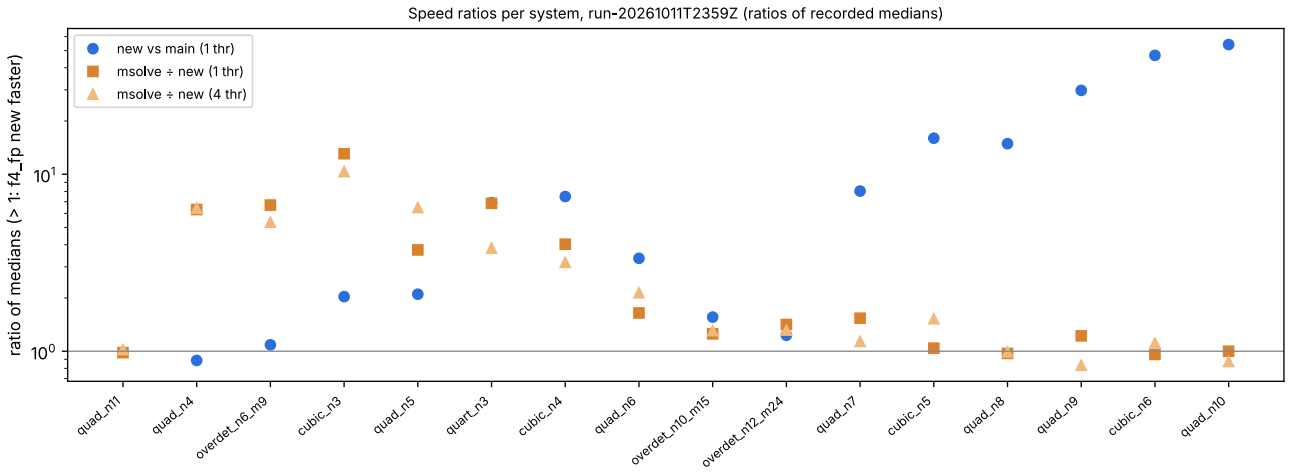


Figure 2: Ratios of the medians above. msolve's ratios on the smallest systems mostly measure its process start-up.

Records

`raw/ours_{main,new}_t{1,4}_{fp,fp-xl}.round{1,2,3}.jsonl`, `raw/external.round{1,2,3}.tsv`, `raw/versions.txt`, `raw/quad_n11_basis_check.txt`. The exported inputs and msolve's bases (83 MB) are not kept: `manifest.sha256` lists their hashes, and `gb_cas_export` and msolve regenerate them. `medians.json` and the graphs come from `../../plot.py`, which reads `raw/` only.